

Kathryn Nave ~ Active Inference Insights 020 ~ Agency, Vision, Life

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hello everyone and welcome back to active inference insights as always I'm your host darus PvZ Wayne and today I'm really excited to be speaking with Kate Nave Kate is an early career research fellow at the University of Edinburgh and her research focuses on developing a realist account of autonomy and agency grounded in the uniquely metabolic existence of living systems she has also just finished writing a book titled a drive to survive the free energy principle and meaning of life which I know the whole active influence Community is looking forward to reading and hearing about Kate welcome to the show thank you so much for coming on it's going to be really fun I hope yeah thanks for having me nice to meet you so there were a couple books as you know about active inference but um they're all quite text Booky and Technical so I'm kind of curious about just generally how did the book idea come about uh and what was it like writing this book and you know overall sure yeah it was I guess quite a securious process it kind of began during my PhD um when I was looking at actually began with looking at predictive processing and visual phenomenology from like kind of a perspective in hilan phenomenology trying to connect that to predictive processing which was interesting but um through that I got

more into the Frege principle and I think I initially began with thinking that the Frege principle was going to be a very useful way to kind of scale up in active accounts of purpose and intentionality and then like get through the course of kind of investigating that it ended up being kind of I guess the book came out of this to do two things for me um the first was to try and get through the kind of philosophical ideas of the Frege principle in a way that doesn't sort of shy away from the formalism but doesn't rely on the formalism if you see what I'm saying because I think it's at the point which I was writing it was well pretty well communicated to people with a mathematical background that wasn't perhaps initially the case um but the kind of communication of the mathematics I think had gotten a lot clearer but it was still the case that if you didn't have any kind of grounding in that there wasn't anything that was accessible to you I think in book form um so the first half of it is really trying to look into what is the what are the what are the formal aspects of the principle actually saying in a way that's explained non-mathematically and intuitively um not to say I mean you know to some people mathematics is intuitive but to some people it's not um and then the second part is I think through that kind of deconstruction of

what is the mathematics actually telling
you it started to become clearer to me
that the kind of aspects that were
supposed to account for purposiveness
and agency weren't really there which I
mean I think now is becoming now maybe
doesn't seem that stranger thing to say
I don't know I think there's been more
papers and more discussion of this
aspect at the time I would say there
wasn't so that was part of it and the
other part of it was sort of discovering
a whole lot of really cool work on the
physics of living
systems that shows how much more
complicated they are than we expect how
difficult they are to pin down um how
sort of pran and changing they are and I
think all that stuff is really cool and
I found that the fu principal is quite a
good launch pad for getting into that
work because it sort of pushes how
far can we go with this sort of
mathematical formulism how far can we
get in our description of leing systems
and for me like butting up against those
limits introduced me to a whole load of
ideas in kind of organicist philosophy
of
biology um in particular that is that's
kind of where the book ends I suppose
with a here's what's kind of Beyond The
Finity principle here's all this stuff
that isn't I don't think for me not
something that can be a part of this
like very nice very simple principle um
and yeah so like kind of like a simple
expl as simple as possible explanation

of the constructs involved in the fry
principle how far can they take us and
what can't they help us with when it
comes to living systems in particular
sure see I I want there's there's loads
of things there um I wanted to pick up
on this thing of you mentioned agency
and kind of Theology and the changing
perception that we have of those Notions
at least within Act of inference Act of
inference might intuitively lend itself
to a kind of realist notion of agency um
just sort of embedded within the idea of
self- evidencing in some way and this
goes you know all the way back to kind
of inactive Notions of autopoesis that
the uh and maybe even earlier to
hiigaran phenomenology that the organism
is taking a stance on its being and that
is self- evidencing that is it's carving
out its own Niche and its own
world um when you say that it's sort of
not there do you mean that it's there
are two kind of ways of interpreting
that one is that it's actually not there
on logically speaking that it's a nice
illusion that we you know create for
ourselves to get by in the world that's
one stance then the other thing is that
it can't really be fleshed out in terms
of the formalisms and there might be
something latent uh in the ontology
which the epistemology can't get us to
at least the epistemology of active
inference so which one is it is it the
uh the instrumental tool of active
inference is insufficient to give us
agency andology or is it uh that it's

kind of an illusion the whole project in
the first place
yeah I think it's very much that the the
the tool that active inference is
doesn't allow us to describe the feature
of reality that underpins this kind of
purposiveness is how I see it um and I
think and I can go into more of like
what those features are but I think
that's why you now have this kind of
perhaps move in active inference to
where previously self-
evidencing um you know self models all
of this was presented as very purposive
I think I'm you're starting to kind of
see more papers that are now saying that
that purposive gloss on active inference
is instrumental yeah that it's useful to
describe certain systems as purposive
and not so useful in the case of other
systems but in all cases all that's
happening is for energy minimization and
perhaps a sort of datian kind of picture
where when you get a certain level of
complexity um it's helpful to use this
kind of intentional gloss but it's just
a gloss um and it's a gloss that picks
up a real difference but it's um a
difference of degree not a difference in
kind I think is the important thing
there so the question of where we draw
the boundary on that difference of
degree is relative to our interests um
and that's not the view I have but I
think that's the view that you get to if
your ontology is limited to what the
frery principle gives you because it
doesn't give you these kind of

differences in kind between what I would think of as agential and purposive systems I living systems and systems that aren't identical and purposive for me there's a difference in kind there um so does this rest on some kind of axiomatic uh presupposition that there is agency and purposiveness embedded within living systems and and where is where's that coming from is it just phenomenology um where where do we how can we ground this kind of starting point that well there is a difference really between me and and a rock let's say yeah I think that's the part of it um so that's not something I get into as much as I'd like to in the book that's kind of what I'm working work on now I think I began with that presupposition because I began with this inactive view of purposiveness and agency and I suppose because I began with working in visual experience I think the way that visual experience is characterized in a lot of the phenomenological accounts that I would draw upon is very purposive and intentional right it's an attempt to there's always talk physically there's very sort of striving notion of perception where it's about kind of bringing things into alignment and achievement and that kind of a very intentional purposeful gloss to it so I definitely began with that very like if you can't everything we want to say about cognition for me is grounded in agency um and if we get rid of that then the

kind of whole framework does sort of
crumble and I think you know people do
feel that you can keep it going with
this um instrumentalist stance I think
there's a lot of problems with that
because I mean one of them is when we're
talking about talk of agency purpose is
being relative to like our interests and
we already we've got interests in the
background there right that we need to
make sense of you know is it a good idea
to speak if this is purposive or not
it's relative to what I'm interested in
how do I explain what I'm interested in
you get that kind of regressive
perspective which is normally stopped
with Evolution I guess it's the standard
move yeah um you go in natural selection
um so there's a reason for
thinking if we don't have agency I think
we lose a lot more than we realize and
there's this gorgeous photo actually
like I love Jerry photo and he's obviously
he's he's very known for his defense of
intentionality and purposiveness in a
very different way from which I'd want
to defend it but he's like if that's
gone then it's basically just the end of
the world he just throws up his hands
and cries um and I think they I do think
there's something right about that like
I think we lose more by letting go of
that it's we don't just lose I'm not
even that interested in Free Will it's
not just that we free will we lose
Notions like function we lose the
ability to sort of demarcate systems
relative to they do it all just becomes

relative to something we don't really understand in terms of section so does the deeper question does the deeper Free Will question really uh feed into your consideration of agency I mean you said there that you're not too fuss about it in some ways it's a worthwhile thing not to be fussed about because it's a it's a philosophical wormhole in and of itself um does that fodorian quote and and your perspective rest or is it contingent in any way on there being free will or is it just kind of this uh parent tellos or this parent function that things have yeah so I think it's very it rests very much on the function aspect not being a parent okay when it being the case that it's not up to you or me what a particular living system is trying to do like if I say any coli is trying to metabolize glucose that is I want to argue um something that's true about that EO like independently of what you or I think about it and I'm quite an anti-realist um so for me that's quite a big move to say there are real facts in the world independent of our perspective and I think that's about the most basic one is these kind of facts about what living systems need those are like going to be yeah the most sort of basic realist facts that you get in your analogy um but the reason I say I'm not that interested in the Free Will aspect is I want to be able to say that certain systems have needs that they are trying to do things that they have things can be better or worse for them

so it's about that kind of there is a sense of responsibility there where we can attribute these things to these systems it's just that the concept of free will I think is so it's one of those Concepts that's almost like a bit um contaminated in that it's very hard to agree what different people mean by it and it plays very different roles in very different discussions so it's not when I say I'm not interested in it I think I'm interested in things that some people might consider as being what free will is about I just don't like to talk about it in terms of will because it's just too messy a concept yeah no I think I think it's absolutely right that we need to just at least be precise about what we mean by the words we use I mean it it ends up becoming a sort of you know people um just just spending half of their papers writing what can you know the meaning of the concepts they're using but a shared language game would be would be nice because I I think um when people hear agency they do think of Free Will and they don't think of this uh this kind of non Lo like for me just this notion that agency is a function performative function uh it's kind of like I don't know for me it doesn't really feel like it's enough yeah but um I also wonder whether if we so we we take function and process as realist real in our ontology do we take uh do you take thingness to be real as well because this is a kind of contentious

Point within an active inference well
maybe not contentious but at least a
point of curiosity which is active
inference

atically on the fact that there is a
thing and that thing pers you know you
can describe a thing as persisting such
that it's the same thing today and
tomorrow is that a viable or reasonable
Axiom or do you think that that can be
sort of

De mystified

deboned yeah just come back to the thing
you said first off about this not being
sufficient for agency I would also the
kind of thing I'm talking about I would
also agree with that I tend to kind of
refer to use the term autonomy for that
reason because I think autonomy is like
a sort of necessary condition of agency
but maybe not the full-fledged thing and
actually before we before we go on to
the second Point by autonomy are you
talking here about a sort of traditional
and activist account of autonomy well I
think where I where I would like Grant
like originate maybe the notion of
autonomy I'm interested in is just this
kind of Canan thing of like being a
system that gives a law to itself yeah
so

that's that plays out in an in an active
concept of autonomy but I think it's a
sort of more thinner or essential notion
that people have developed in these
different ways but the basic thing is
how can you be a system that constrains
yourself because you know if you're the

only constraint on yourself and surely
you can just choose to stop constraining
yourself um like it's a really to me
it's a really really
lovely philosophical kind of um I guess
an antimony in C or like a paradox where
there's something really important about
the idea of self-c constraint it seems
like a self-defeating
idea and that's maybe that's kind of
what I mean by autonomy is like this
really all of the confusion about agency
and all of the reasons that agency in
general I think is difficult to explain
is because it's for me I think it's
Central ingredient is this kind of
autonomy of being a system who has
things that you have to do because of
the system that you are that aren't
externally imposed so it is a very
active Nation as well they're not
internally imposed needs but leads you
still have to satisfy them so what so
what what kind of example could you give
there so for examp I think obviously
that se that additional Clause is
necessary because then I can say well
the rock is autonomous because it has
needs at least uh you know from the
mathematical statistical instrumental
standpoint I.E to minimize variational
free energy what needs could you say uh
are do we impose on ourselves which are
not I mean I'm I to put lay my cards on
the table I'm personally quite quite
skeptical of the internal external
boundary in any case I'm not sure what
it means to have a internal versus

external constraints because underlying all of that you need to have some locus of free energy minimization right so you have say it's the dorsal horn or you know Andy Clark might say it's me and my phone and so on but in any case uh I'm I'm quite happy to have a kind of process ontology about where those boundaries lie um but what is it so what would it mean I mean if we can say the rock clearly doesn't have kind of this interiority whereby it's uh creating its own norms and it's abiding by those Norms that I seem to be um can you give us an example of like one of those because for me I feel like I can always reduce it back to some kind of exteriority uh sure like so the way I think about this kind of self law is is then does become a more an activist notion though so that this the kind of framework I particularly like for it is um the kind of biological autonomy framework that's been developed by um alv Marino maffei mosio in terms of constraint closure which is very inactivist but I wouldn't necessarily say they're inactivist so in that sense um but the the way the way that you have a system that has something like an interiority exteriority distinction but not obviously self what's it self like not self-standing completely where you have that kind of operation that kind of constraint closure which is their development of what would be organizational or

operational closure in an activism but
you know you have all these different
versions of closure that are all
actually really quite different
um but sorry just a second to come back
to it so the idea is the way in which
that gives you this kind of need that is
your own need that a rock doesn't have
is I think in two ways firstly there's
the sense in which I think when we speak
of non-living things those are things
those are carvings out of the world that
are relative to our interests so
something like a rock for me is not a
natural kind um anything that's not
living arguably I would say those are
kind of distinctions that we're placing
on the world we're saying to be a rock
you need to not be heated over a certain
temperature because then you will
collapse and no longer be a rock but
that's not really got anything to do
with any sort of self- individuating
system um that's just my definition of
what a rock is so to say that you know
you have you can have this sort of
statistical model of a rock and then you
can say that for energy must be
minimized relative to that model if it's
to stay as a rock that's still very much
like my imposition on the world when you
have something that's I guess a
metabolic system I think what you have
is a system where there are needs
intrinsic to that Network where each
part of it depends depends on other
parts in such a way that it's not
about how to put this how this so that

there's a stability but the stability
depends upon the activity of the system
so if you just leave it on its own um it
will if

you I always find this really difficult
to talk about if you it's kind of like
the it's kind of like I think the the
classic example is something like the
cell yeah then the cell has this uh
partially permeable membrane and it
needs the sorts of metabolic resources
in which to continue to generate that
partially permeable membrane I mean I
think that's a classic autopoietic notion
the thing is I'm being really annoying
but why not it's philosophy um I just
wonder whether that kind of so that
thing of uh in terms of operational
closure the individual constituent Parts
rely on one another so without one you
don't have the functioning of the the
whole

and and I think you know they term that
precariousness um I just I wonder
whether you could consider the in if
there is the interiority of a rock to
also be precarious in that way such that
if the central bit of a
rock is heated to a certain degree such
that it melts the whole thing melts
right in some sense the whole thing
collapses and become soil or at least
not the singular rock that we thought it
was it would become a multiplicity of
rocks I it's interesting that you carve
out the sort of natural kind and and the
rock is not necessarily A distinct thing
in and of itself and it's an imposition

I kind of in maybe it's a very uh
exaggerated Buddhist idea that I think
we're the same right I think that I
think it's uh it's difficult to say that
I am
ultimately physically distinct from the
chair on which I'm
sitting now I think the one I I spoken
to Carl about this and I think the the
what he rests on is the idea of
action uh and I guess what in an
activism would be adaptivity so there
kind of Ezekiel dealo idea
that by because we can take a stance our
own being we can change the world in
accordance to our prior so to use active
inference language how does that
solution stand for you I mean I I you
know just to sort of tee you off is
there an issue with the underlying
notion of action as well is that is that
a Troublesome concept yeah um so I
think it's funny like when you have this
discussion about whether or not living
system is a a self- individuating system
or not that you you're either often
talking to somebody who would just
absolutely accept that rocks and chairs
and things are just real things in the
world independently of us or you're
talking to somebody from your position
on the Other Extreme of it and I find I
suppose I find the the end of the
spectrum that you're more sympathetic
that's where I would have initially gone
lent before I was more convinced by the
this idea of self- individuation and I
think yeah so I think action is very

pivotal but I think it's often used in a very weak or general way in which I think we can make a distinction between an action and a movement and if you just refer to all movements as actions then you're not going to get any sort of distinction between an operationally closed system and a rock because you can say that a rock depends upon all the forces that all of its different particles exert on each other to hold it together you can call those intermolecular forces actions right can say there's a network of actions between the various components of a rock keeping that rock going yeah yeah yeah so yeah this is helpful this is kind of what I was trying to get to before was that why is an operationally closed Network different from a system where the Integrity of the whole does depend on interactions between its parts we can think of a rock that way and I think it rests on having this notion a very I guess quite unusual but quite stringent notion of like what it is for something to be acting on its part where an action and this is what you get in the kind of constraint closure formulation is an action is when you're constraining a release of energy to perform work in a way which is General again like um you know a channel dug into the mountain channeling water down a hill is constraining the release of energy to perform work of turning a water wheel so the idea of constraining

a release of energy to perform work is not sufficient but it becomes an action where when the constraint itself is dependent upon the work that it does when you have that kind of circularity so and that's what you do have in metabolism of course because in metabolism you have these enzymes that are catalyzing reactions that only happen because of the enzymes but if the reactions don't happen the enzymes themselves aren't regenerated and that's a very different relationship between Parts than you have in something like a rock and that's where I think you do start to get this more genuinely self-individuating thing and it's why I'm not super key on this kind of way in which precariousness is often talked about in the inactive literature because precariousness can sound very vague you know everything is precarious over time everything will collapse over some period of time it has to be a very particular kind of precariousness that is distinct to living systems which is I think this very particular kind of closure great no that's that's really useful um you mentioned constraints there a couple of times yeah um and I feel like there's a this is not a literature I'm super familiar with I know work of someone like alicio jarero and this notion this kind of Distinction Aristotelian distinction I think between causality and constraint in your own words what what do you understand as the difference

between those is there again is this a natural you know a natural difference of natural kinds that there is something genuinely ontologically different in the universe between causality and constraints uh are they the same thing with different you know words attached to them um what how do you understand them and also what does that distinction give us in terms of our scientific or philosophical Outlook yeah so I think one of the caveats here is I'm I've not done a whole load on the metaphysics of causation as it's kind of generally understood and I think what I love about constraint is constraints make sense to me when causation is talked about in a lot of the philosophical literature at least my sort of surface level of awareness of it is it gets very messy and you end up with views on which there's no such thing as causality because if you have a particular notion of causality you don't find that at the level of fundamental physics um and then you have I think among sort of philosophers maybe more like myself who are in something like cognitive science or not in the kind of philosophy of physics but in more General philosophy of science areas you can sometimes have like a very sort of traditional basic notion of like billiard ball causation an event happens it triggers another event and that's just not very useful for a lot of the kind of things you to talk about um whereas I think constraints are particularly for the kind of causation

so I think constraints are a type of cause and for me they're the type of cause that makes the most sense um and they are the type of cause that helps you talk about living systems really well because they get that element of the way in which it's very difficult to talk about interiority and exteriority in the living system because there's such a dependence on exteriority there is also some sort of closure and what the idea of constraint causation gets across is that dependence on exteriority because you need the you know you need the input of energy to be channeled by the constraint so it depends on something from the outside to do anything but the thing itself is what makes things happen and it gives you that I think also that sense of sorry uh being able to identify a cause because if you just have the billiard ball thing then you just Trace back that chain of events at the beginning of the universe and it's who started it whereas if you have a constraint plus even just like a random um spontaneous reaction you have two things that you don't need to go back on a causal chain like the event of a reaction happening only happened because this particular constraint was here and this spontaneous release of energy occurred so you have that sense in which it's much easier to identify I think it's much easier to pin down a cause when you think of it in terms of constrained releases of energy

rather than events triggering other events although I can imagine a very picity philosophy of you know philosopher of science would say that you could also trace the constraint ad infinitum right or or add beginning of the universe because uh presumably you you know constraints presumably have some kind of like a fractal nature such that something is is constraining only because it was you know the result of some other constraint which in of itself was a result of some other constraint uh I mean that's just addendum I think I think they're both useful ideas um I don't know my experience is that we're as you say stuck very much still in the kind of humean um skepticism which I think is totally reasonable I mean I have no I have no problem with Hume but um yeah I think you're right constraint certainly gives us something else does how does constraint therefore fit into active inference because we talk a lot in active inference of hidden causes and latent causes um you know one or the other uh which implies some kind of underlying acceptance or Axiom that things cause other things to happen right and that's in kind of inherent in uh generative modeling or basian design that there's quality is again that could we just deflate that whole thing just by adopting a kind of humean skeptical attitude such that we can kind of keep these words but we recognize they're just labels for something that we can't necessarily pin down ontologically or

should we actually be discussing constraints and does that give us anything over and above uh the latent cause hidden cause out there sure yeah so I think in terms of embracing the hean skepticism thing that's what I don't necessarily talk about completing those terms but one of the things I kind of tried to go into in the book it's just in an appendix because it became kind of a degression was like what would it mean to kind of accept because there is I think you know one of the things that i' gets criticized for a fair bit is this conflation between statistical and causal Concepts um so you know to take a a whole L of statistical relationships and say this is a causal graph and these are like causal relationships and these are real boundaries and all that kind of thing and I felt I feel like sometimes that criticism is actually a bit potentially unfair in the sense that you know there are plenty of philosophers who going back to him who have defended that kind of stat what you could call a statistical reduction of causation and know there's tons of work in like Cal inference where people have tried to look at how how those are related um and you know there's there are there are philosophical World Views in which the world is just a series of you know SpaceTime points that stand in certain statistical relationships to each other and that's all the causality is um so I

think that would be a I think that's
like an interesting direction to go with
active inference um it's not the
direction I would go but it is like
there is whole lot that could be said
about like a particular metaphysical
causation that would be really
compatible with the kind of active
inference approach and make that
distinction between the model and the
territory if the territory was that way
a lot smaller um yes yes exactly no
that's absolutely right it's kind of
it's nice because I don't agree with
that but I think it's like an really
interesting position that could be
developed yeah well as you say I think
uh well it's it's a life to do
metaphysics cation is a lifetimes
project I think um
so we need some philosophers who have
done that since they've been sort of 11
I feel like it's one of those things
where you know yeah you devote a whole
career to the metaphysics of causation
and then hopefully you apply it to
something like active inference that's a
really interesting point that if we got
a concrete handle on what it means for
something to be a latent cause and it's
not just this um well in perhaps an
instrumental conflation of uh
statistical cause and and and an
ontological and genuine cause we could
collapse somewhat the map territory
distinction is there it's just the
question comes to mind is there anything
else that you can imagine if we got a

sort of proper grip on or if we said
okay this is not just instrumental it's
it's a genuine Claim about what there is
in the universe we would uh we would get
closer to reducing the uh map territory
distinction is there anything else I
mean I sense agency is one of those
things as well right um because we could
claim that it's a kind of uh more like a
map feature rather than a genuine
territory feature if you're thinking
about the the Deep metaphysics of Free
Will is there anything else that kind of
came up that you thought at least
there's a question here of whether it's
a it's a map or or or or whether it map
belongs to the map or the
territory just to I for me the agency
belongs very much to the territory I do
think it's a part of the territory that
is kind of missing in the for enery
principles map yeah
yeah in terms
of I think the biggest move would be to
argue for a statistical iCal reduction
of quation I so like that's not another
thing but I think once you've argued for
that then other aspects of the fry
principle become more
realist
um what do you think I mean uh one thing
that I think about a lot because it's
kind of what I predominantly focus on
and write about is the
self um so some people take the notion
of well it depends on kind of your
underlying ontology but some people do
take the the notion of a kind of I think

I kind of think that cartesianism is having a bit of Revival um I'm not saying that I subscribe to that but other you know other people might and I think in some ways it's it's aided perhaps implicitly by a framework like active inference because it very much forefronts the the self- evidencing agent and I it came to mind and I was going to ask it when we were talking about interiority versus exteriority because um but you know the conversation moved on but it came to mind that are we always at risk of uh implicitly invoking aelf when that would be uh a misplaced um addition to the conversation right like is the distinction between myself and the rock well perhaps implicitly and intuitively I think it's well I'm I have a and The Rock doesn't now this is relying on very strong uh dualistic ideas of what what self is I wonder whether getting stronger on what it means for self- evidencing like is it purely a phenomenological claim that we have a sense of self is there something underneath the pure statistical mechanics and the free energy minimization I just feel like that would get us a lot closer to a a more realistic account have you been thinking about the self in any yeah so I think the way I'm thinking about the self the idea of self a self that distinguish disting self that distinguishes itself um is very tied up with me for agency with agency so I think those two things

are kind of are part of a package

because the way I think about agencies

in terms of autonomy in terms of giving

a law to oneself all of that kind of

um

so and perhaps this is

where I think when you take the frity

principle as metaphysics and you take it

you take something like self- evidencing

as being what gives you the boundaries

of the self I think what you end up

there my impression of it is more like

kind of a position that perhaps you

would be more sympathetic to a position

in which there are no real selves

because everything can be kind of not

everything but self- evidencing is kind

of everywhere right um if if causation

and inference are understood in this

very very minimal way where all it is is

to have the right kind of statistical

relationships to the things around you

then you going to have Mark of blankets

all over the place you're going to have

the ability to describe one thing as

modeling another all over the

place if that's all that you take the

self to be then you're either it seems

to me going to end up with a very

psychist view of s or a view in which

well if s are everywhere then there's no

not really any such thing yeah I think

that's reasonable I mean I I maybe I am

lending myself towards pan sism because

I do think that um phenomenal experience

is sort of the one indubitable fact

about the about the world so uh that's

just that's that's kind of we can I mean

I want you know we will get to ious I'm
sure but without without going there yet
and perhaps it will come up now
is if I adopt if you don't adopt that
kind of deflationary account of
selfhood what is there in your ontology
that's over and above that so we spoke
about agency and how it was about sort
of a law onto oneself and uh autonomy
and that would okay there we kind of
stake the claim about ontolog or you
know on actually what's you know what
might distinguish me
morck is there anything beyond
Consciousness it might just be
Consciousness that in a sense adds to
this notion that I might have a self or
might be a self whereas a rock isn't
what is that over and above um this very
deflationary account that I'm proposing
or hypothetically proposing yeah I mean
so I think one of the things you
mentioned earlier is like the role in
which constraints play in the for princi
and the role that constraints play in
pretty much like I'm I'm not a
mathematician um so I'm sure but in
pretty much every mathematical model you
set up some higher level constraint yeah
then you which is expressed in the form
of invariant equation and then you can
have you know lower order equations and
parameters that change relative to that
higher order equation but at some level
you're always going to have that stable
higher order equation and that's
basically a constraint like an equ you
know the kind

of in so I don't know if you've have you
have you ever come across Robert Rosen
at all I don't think so okay um so
basically he was a I guess theoretical
biologist um and had this description of
living systems as systems which generate
their own equations they generate their
own functions nice um so you have you
have two things essentially you've got
your variables and your functions and
your functions are for him God I'm
terrible on my four causes your
functions are your officing causes and
your variables are your material causes
so your variables are the stuff that
goes in um your functions are the things
that does the thing that does something
with that stuff um he calls them
efficient cuses I think there's an
argument they might be more like your
formal cause your structure um whether
they're formal or efficient is I think
an interesting discussion um but the
idea there is that what you have in the
living system is a system where you
don't have that higher order independent
equation that sort of sits on top and
doesn't change instead you have this
cycle where every sort of function
making up that system is modified by
some other part of the
system is that that that's okay that's
an interest so well okay so that I feel
like in some ways comes to this
deflationary account of the self right
there are just what verella called
selfless selves without some
superordinate homunculus that's that's

guiding the show um okay that's but do
you

get what what what is what what's left
in the concept of self if that's the
case if everything is just precare if
the self is precarious in the that there
are a multiplicity of selves that rely
on uh one or or the rest for the for its
own

existence can we be left with any cogent
concept of a self at least from what the
word we think means yeah I mean so I
think you don't you definitely don't get
like a cartesian self you don't get a
lot of the stuff that people pack into
the self I think you get you get for
example an ability to be there a way of
deciding if something is part of me or
not a part of me um so that's what you
get with the cell right the cell lets
things in it doesn't let other things
in and that doesn't mean that something
is that is not a part of me or not a
part of my cell can't be a part of a
broader thing that we're both part of so
like you know a cell in my finger is not
a part of a cell in my eye right they're
both part of a bigger self that's me um
but with

the like this kind of
recursive uh what for some reason called
the Mr closure but this kind of closure
functions what you don't get is anything
stable I think that's the really
interesting thing that's different about
this notion of self you get that kind of
individuating network where everything
is defined in relation to something else

and some things are just not part of it
because they don't enter either as a
flow of energy into the network of
constraints so functions of
constraints um functions of your
constraints and the variables changing
or the input of
energy that does allow you to demarcate
a sort of cycle that other things are
clearly not part of because they not
generated by it nor do they enter into
it but what it doesn't give you is any
particular thing that has to stay the
same yeah yeah apart from what does the
boundary have to stay the same not
necess there has to be a boundary but
there's no reason why the boundary
couldn't change right so do we then say
I mean this is a very sort of
caterpillar butterfly problem or or you
know so there are some great work you
know I spoke to Mike Le
uh about a month ago on this podcast and
Andy Clark has got some really nice
papers on this kind of process ontology
perspective um are we then
invoking uh like if we just take the
lifespan as our sort of time constraint
our time boundary are we invoking
multiple selves then or a kind of
superordinate system to which this
metamorphosing um multiplicity belongs
is it is it I mean this is going to come
to this conversation about really like
is that can we if we keep recursively
building out a level bigger levels of
abstraction and gen generality is there
some underpinning rule is there some

underpinning guide to that process
ontology such that I can point at
something in the universe over the span
of a lifetime and say that was uh Kate
or that was Darius or or or we
just you know are we just actually just
multip are we just multiple multiples
you know I think sort of both yeah
I think one thing that kind of annoys me
with pris ontology sometimes is I think
it's a bit
too you know the selves all over the
place this could be a self relative to
some perspective but this could be a
self relative to another perspective so
what I like from the kind of closure
view is you do get it if we're looking
at this
scale then this is the self and this
isn't like that's not relative that's
not quite as pluralistic as a the more
pluralistic ends of the process ontology
um but it's always going to be the case
that if you take again the idea of
constraints sorry was talk uh as you
should um but what counts as a
constraint is always defined relative to
some other process so on sometime scale
this is going to be an invariant thing
that's not changing but channeling a
change but is invariant itself but on
another time scale that's going to be
changing and going to be need to be
replenished by some other activity um so
those things are always defined in
relation to other things what they're
not defined in relation to is my
interests as an

[Music]

observer so you do get this sense of
yeah multiple levels of
selfhood but you also get this sense
that once you fixed a particular time
scale what is going to be a self or a
particular space what is going to be a
self under that is not relative or
variable yeah
yeah um I'm just wondering
whether if so Rosen is a biologist I'm
wondering whether is is there is there a
limit to like over that time scale sure
it's not uh it's not boundless there are
limited forms that it can take it the
constraints of that it sounds to me I
mean the biologist will say this sounds
to me like the actual physical body is
very much one of those things right um
now I know people will say well the
cells in your body change every however
you know however often in in the course
of a lifetime but um you know you say
the boundary can change to what degree
are we taking the the epip helium or
wherever you want to carve your boundary
to be a
significant um self- evidencing if you
want to use that term sort of
limit yeah so I think the
interesting the
interesting it's what this I think in
all of these things you have this really
interesting tension between there's a
sense in which it can change and there's
a sense in which it can't
change yeah yeah yeah but that at each
moment I would say the current

configuration of the system is pretty
constraining on what it can do right so
in each moment what can happen next in
the very next instant is relatively
limited yeah but how that Pro as with
Evolution and you know how that can
unfold over time is entirely
unpredictable right because it can open
up because these these changes aren't
just I suppose quantitative it's because
sorry which quantitative qualitative
because it's not just say an increase in
size it's not it's not just that the
size can only go up by this much each
time it's that once the size goes up to
this much a whole other range of
possibilities become available like so
you get these discontinuous sorts of
change yeah at point if you follow it
through time the possibilities will be
small over time they just
expand um beyond measure so like there's
this very nice notion that Stuart calman
has is like the adjacent possible so
this point maybe your possibility space
is relatively tractable but because as
you move into one area of It new areas
open up you can't just constrain
yourself to one area of possibility
space in that way for a living system so
I guess time yeah so the the the
spectrum of time is really very
important here in terms of the
possibility space of constraints um I
found it interesting you said it's not
predictable because I understand that uh
there are two ways you can kind of frame
that one is is not predictable for me as

an energy limited human being scientist
uh and that's because I'm energy bound
um and as you say once time gets beyond
the $t + 1$ one it becomes combinator
explosive uh as to the possible State
spaces that I can be in okay that's one
way of putting it the other way saying
it's not predictable is that it's not
predictable for $t + 1$'s demon I it's not
predictable uh in the sense that there
are genuinely different
options for like there is it's not the
this the system really could have done
otherwise right that's the classic
phrase as you know in the sort
metaphysical Free Will so which one of
you kind of pointing to that is it
genuinely the fact that because of the
combinatorial explosivity of State
spaces um it's unpredictable for me
which is fine it's unpredictable for me
what you're going to do next anyway so I
mean okay that just shows my limits or
is it genuinely the case that there is
not there is more than one path that A
system can go down and therefore even
the best you know infinitely knowing
mathematician would be unable to write
down the path integral that describes
the
uh you know movement over the time the
DI chronicity of this being yeah so the
second argument is something I'm
starting to think about but
I I think you you just have to be so
careful in terms of arguing that they
are genuinely random events I think what
you have with living systems is

certainly the case that you can have events at a level small enough so they could be genuinely random can have a kind of effect that leads to a combinatorial explosion um and that's very distinct to lyricism so you know even with like you know you get um macroscale events from like microscale events even just in computers right which is really cool like you can have these single event effects where random particle emissions flip a bit or something and airplane crashes right which is super cool it's super cool that you can have that kind of macro scale causation from something as small as a photon right and that's just that's solely because of you know the incredibly small size of modern computer components but what you don't have is that having ongoing repercussions that's what you could have in living systems where something like that scale of event not only causes something to crash or something microscale to happen but then opens up a new region of possibility space then opens up a new region and like you get combination yeah I guess the question there would be I mean there are two questions there one is is it not the case that something like a human would have an attractor set to which is almost uh again I'm not invoking any deep teos here but it's almost responsive to those perturbations right that's kind of the classic vellian

response is that all we have are perturbations and uh we have these I mean he probably wouldn't he didn't turn in this way because dynamical systems theory came after the kind of autois but we have this return to this attractor set such that I have this itinerant as Carl would say movement towards the attract set but that doesn't mean I'm always in that possibil like that high possibility that peak of the gaussian I can I can sort of be pushed away but I'm always returning up the hill in some way that's one question and then added to that at a more General level even if uh you know someone might come along and say sure you have these random Black Swan events and uh they have these trickle effects such that it's impossible to predict where and you know t plus 10 I'm going to end up because combinatorial explosivity someone might come along and say yeah but there's actually just a even bigger attractor set which governs the transition of you moving into this place which seems random but actually like the photon ing the plane it was always going to happen and actually it was in the plane's ontology that it had that attractor set of ending up in the bottom of the ocean yeah um so I think this comes back so the reason I haven't focused on the randomness aspect so much is precisely for the reasons that you've just been talking about is I think that's the less important component the more important component for me is the sense in which I

don't think I think the youve reach a point where to continue arguing that there has to be some sort of stable attractor for that can explain everything that I do it becomes so General that the selves which is the self the the selves kind of disappear under that because this is an attractor that this is a sort of description of invariant behavior that we all share so we all you know have to couple anabolic and catabolic reactions in the right way to keep our metabolism going that's a very general principle that we'll always preserve um but that doesn't give you any sort of individuation that doesn't give you a sense in which you're different from me and we're all different organisms so you can have very general invariants but those General invariants don't allow you to pick out individuals or you can have individuals but those individuals I think don't have invariance and the way that that as I understand it tends to be handled in the free energy principle is you kind of go to this sort of middle level so if you're looking at what's the invariant you know very simple example um if you take something like a EOL bacterium and it has that ability to switch its metabolic system you know from glucose to lactose and you can say well you know what's the stable model for that ecoli because up to this point it's only metabolized glucose and now all of a sudden it's doing something

completely different that's a phase transition and its behavior and energy principal says it should die you just go oh well no that's okay because there's a higher level model for a colite in general at the level of the species and according to that higher level species model it's stable yeah exactly yeah and I think the Reas I think that doesn't work and I think the reason it doesn't work is because individuals can also persist through transformations in their behavior that are novel at the level of the species um so you know you can have you have the really cool examples of metabolic plasticity and cases where things like bacteria do like swap their DNA and gain new abilities and they might gain at the point of which that happens there's going to be an individual who gains an ability then no individual of that species has ever had before it's truly novel it can still be good for that individual and it's still part of that species presumably well yes we want to say we want to say it's the same individual yes and it's part of the same species but what's occurred has never happened before at any level of that individual's kind of generative model so is the so under this is the butter is the butterfly the same as the caterpillar yeah yeah yeah Mike said this really amazing thing which is that so butterflies I'm curious about what you're think about this butterflies and caterpillars eat

different things right so caterpillars
like leaves and butterflies like nectar
but if you have so you can uh through
sort of very low level conditioning you
can sort of make a caterpillar go to a
specific Leaf of a specific color
because you've trained it there's food
there's going to be food there and what
he said to me I haven't actually looked
at this is that uh you it when it
becomes a butterfly it it stat
statistically goes back to that leaf
more than by chance which implies that
despite the and so the kind of big
question there is where's the memory
right if you're a neuroscientist you're
saying well uh I don't know what the
size of a caterpillar's hippocampus is
but it's clearly it's completely changed
right the architecture is completely
different and yet the cognitive function
or the output of that cognitive function
seems to be the same um I'm not asking
you to solve that

Paradox don't worry uh uh but what I am
wondering is does the free energy
principle solution just to say well just
broaden out the Markov blanket right and
just say well we didn't know right we
didn't know that the the eoli was going
to become lactose metabolizing we didn't
know that the caterpillar was going to
become the butterfly does that make it
does that reduce its import right like
is there a problem this is kind of a
meta question is there a problem with
something that's just an hawk or post
Hawk

description um and if it is just that
does it really I mean what what does the
free energy principle say if it's just a
post hoc or ad hoc uh description yeah
um so I mean is there a problem with it
it depends what you're doing um if
you're trying to build an ontology of
living systems then I think it is a
problem um if you're trying
to describe what living systems in your
experience have tended to do in a like
succinct and convenient way then it's
not a problem at all um and I think you
this is a thing that comes up when I
make these arguments about how living
systems can change in all of these
completely not necessarily completely
unpredictable but completely so there's
kind of like a neologism that St is like
unpre statable in the sense we can't
write down which people hate and I but I
I I it's useful it's useful you don't
want to say predictable what I mean by
unpre statable is we can't write down
equations in advance for that individual
so it might be that all of the equations
of the entire universe would allow us to
predict it but there's not one
particular set one particular
mathematical model for that individual
that we can write down that won't change
um God I completely forgot my third was
the No it's it's was that oh uh it was
it was just what oh well the real
question was what does the free energy
principle say like okay it's well I know
what it says right I know it says that
things look as if they are conducting

variational basian inference over you
know their internal States look like
doing that over external States but
really what philosophically speaking
what does it say about things as they
are in the universe if it's just this
post hoc description of not only this
not only like even if you have the path
integral description of free energy
principle I'll give you both the state
space and the path integral for
formulation if it's all only saying okay
I'm just describing how things uh you
know the attractive set that they tend
towards and then also the path that they
tend towards what does it actually say
and I don't mean here in terms of
utility right we're being philosophical
I don't care about whether it can build
robots or not I'm asking just
philosophically does it actually tell us
anything ontologically about what is
going on in the
world so I no I don't think the FR
principle does to be honest
um but I think it does raise like an
interesting problem for the
I think it's a utility kind of argument
but I think the fact that it does seem
to work quite well for allowing us to
describe lots of things does raise an
interesting problem for this view that
if living systems don't have to stay
stable if they can change in all of
these if they can completely change
their entire structuring
configuration why don't they do it more
often right because they do tend to be

stable like right so when I make these arguments about that kind of unpre stability you can look at me and be like yes but we are homeostatic you know we do tend to stay very stable our body temperature is pretty consistent um and that's

where what I've been really concerned with is the fact that that's not necessary it doesn't have to be the case that we have to be stable in that way to continue existing it's really interesting that we can identify something that continues to exist if it isn't stable so it's not a necessary principle on an individual system that makes it even more interesting to ask given that it's not necessary why is it so common um and

I think the position I sort of end up at is you have this kind of system that can change in all these unpre stable ways that that would make it sound like the frey principle isn't going to be useful for allowing you to describe that system clearly it is and I think it is because there is something right about the FR energy principle in the sense we are trying to keep ourselves stable I think it's right that we're trying to do that a lot of the time that we are trying to resist new disruptions not always but that is a part of what we do and a part of the reason we do that is just this kind of suppose like cian gamble in the sense that this has worked for me in the past sorry hum not human will probably work for me in the

future because that's the best we can do
not because that's a necessary
requirement like it's an approximate
strategy to what we're really trying to
do which is to keep this metabolic cycle
going yes yeah there are lots of things
that come up there I mean one is uh that
I'm probably a little bit more skeptical
about than you about the kind of uh
imperative aspect of this um that humans
are actually trying to do anything at
all uh at least they might be but I
don't think the free energy principle
gives you that yeah I think the free
energy principle gives you like if there
is one if there is one human at this
point it's the same human the next day
what do they look as if they've done and
again that's like hyper deflationary but
that's genuinely all I think it does
give you um you would need something
like evolution by natural selection or
some like some more processed Theory um
rather than the principle I think to
give you what humans are doing right
because as you say um we don't need to
do this right like I don't like there's
no there doesn't seem to be an
prescribed law of the universe which
says that I need to persist which makes
it quite remarkable that I do and I
think uh I haven't really thought about
this enough because I'm not a physicist
but I think this is beyond the pay grade
of the free energy principle uh which is
why things abide by they do they look as
if well they do in some very inflation
away but why they do I think there are

some arguments I think Carlo Celli has
some arguments about why there are these
neg entropy pockets in otherwise
increasingly entropic Universe um I
have to speak to him about this um but
yeah it's it's fascinating I mean I mean
again I won't go back to it because I
think we've covered this a lot but I
think the fact that you said that the SE
the the self whatever that is can
persist uh even if we disrupt this kind
of attract attractor set idea like even
if things don't tend back to equilibrium
that the same thing can persist I think
is a really interesting idea um I don't
know how much I agree with it only
because I don't know what if we're
talking about the same thing when we
talk about the self or the same thing or
what equilibrium really is I completely
agree that uh there is no such thing as
perfect equilibrium right I mean I mean
in of itself we need information
transfer to act to for there to be
anything right if not it's just well
there's nothing right things need to be
changing
um what that means in terms of
identity um I need to read the book I
have to preface this by saying I haven't
read the book and I will read the book
but I haven't read the book um but I
what I have read and what I want to go
on to your more shorter although some
are quite long uh philosophical papers
uhuh uh and the the articles so I
thought they were really cool really
really um

insightful uh and and yeah very very
thought-provoking and you as you say you
focused sort of before the book on
perceptual phenomenology or or or what
kind of more specifically visual
experience and sort of probing this
question of to what degree is perception
enslaved by action right like perceptual
experience as is as can be attested in
the phenomenal ology uh why does it
appear as it is and how much has that to
do with action so I guess the first
question I'll ask you very generally for
the audience although I kind of have a
notion because I've read the papers is
to what degree is perception the slave
of action uh and to what degree is it
not yeah so I think my thinking on this
is very strongly influenced by um philos
called Susan Hurley who very early an
activist um well she doesn't describe
herself as an activist but she's I think
meet

I think she yeah I think she absolutely
makes sense to be described as an
activist in in quite a narrow use of
that term so not just
anti-representational um but as in being
interested in the dependence of
acceptational action and her kind of big
focus is on something that she calls the
myth of the giving well sorry it's not
her big that's one of her many focuses
um but so the myth of the given is the
you know this idea that we just receive
conceptual content from the world in
perception and a way of avoiding that is
to go well no you know our experience is

constructed relative to our actions but then you're presuming that the content of an action is already known that you have an account of what it is for an action to have a goal I think this move to say perception is for Action is sort of correct but it doesn't really solve anything because then what is action for um and too often people too often I feel people sort of stop at this perceptionist for action and look that allows us to explain a lot of things about perception but they're already they're kind of presuming a concept of action and what it is for an action to have a purpose isn't already that well understood so yes perception is for action action is also for perception you know it's a it's a circular dependence yes um but what is this whole circular dependence of action and perception itself in service of is I think the interesting question that kind of lies behind that well active inference gives you a nice answer right uh that it's well does it uh maybe predicted processing does actually yeah actually let me ask you does active inference I predictive processing I guess does because it has some underlying teleology that it's about prediction error minimization yeah case I mean you know it says how things seeming you know how the brain seemingly minimizes prediction error when you say uh why does why is uh why are perception and action coupled in this interesting way and you talk about sensory motor contingencies and whatever

and the answer from active inference
person would be like well it's all in
the game of reducing variational free
energy is that a satisfying answer for
you no good good because I think you can
you know you can say all sorts of things
are in the service of something else but
for that thing in itself to be a purpose
process that's very difficult to make
sense of right um and there's a really
nice paper by I think it's Tony Cho and
gosh I forgotten his first name after
for Michael Anderson yeah there's a
Michael Anderson y terrible this is the
problem with only reading people's
papers I know I know about it yeah and
it's like really short but they just
make this distinction between like
Notions of predicting like two Notions
of prediction so you can have one thing
is predictive of something else a
predictive of Rin um and if you just use
prediction in that sense then you can
say everything is in the service of
minimizing prediction error because
everything is in the service of being
correlated with what it is in fact
correlated with right so if something
happens the thing that is most likely to
happen after it is particularly likely
to happen it's incredibly deflationary
boring incredibly deflationary so you
can have that notion of prediction and
everything is being prediction
minimizing that does allow you to
describe all sorts of things as
prediction minimizing which arguably
what the free energy Principle as just

as an addendum arguably the limit of the free energy principle um yeah so I think if all you're trying to describe a predict what you can call predictive relationships all very well and good but a predictive relationship is not a prediction yeah it's not a predictor um so to say that everything is in the service of minimizing free energy or minimizing prediction error there needs to be an account of what makes the minimization of prediction error a purposive process as opposed to just I look at these two things as predictive of one another and see that they're correlated what makes it an attempt or you know what makes it a goal directed prediction that's what's not in the free energy principle it's not what's that's what's not in prediction I've processing there we either go deflationary or we presume that there's an agential and predict predictor a predicting process going on it's not really predictive um but what makes it a purposeful prediction is exactly what isn't in this framework I think yeah okay cool yeah I mean arguably I mean I'm not going to I actually have no bone to pick here with any of this I'm happy to leave it all open but arguably it's because there isn't any purposiveness or agency but anyway you actually took my question way deeper than I was going to than what I meant by it which was that um but I like that it's a really it's a nice add on to the conversation we've actually just been having um I meant more that uh you

write and I think a lot of this is influenced also by Andy and Andy's work and and it's almost a response to Andy's work which is that perception in many ways um is nothing more than the process that kind of puts you in contact with action by that I mean that uh we perceive the world in the certain way because it licenses us to act on the world and I've been kind of thinking this almost in terms of Consciousness more generally maybe what Consciousness is is kind of course graining filter that permit action upon those phenomenal contents and here you talk about um the intermediate level representations or intermediate level processing so what did you what what what is intermediate level processing what are the kind of claims of that uh position that theory yeah I think so I think trying to remember who actually I think it's Jesse Prince and Frank Jackal for they yeah they're the kind of original intermediate level people um and what really nice about predictive processing is obviously well anything hierarchical um is that you don't have to have just one intermediate level there's ler levels you can Mo up and down them um what the intermediate level idea gets at is this idea that what we as agents act upon and not you know individual sub subatomic particles um yeah you know they're not we don't act at that level and we don't act at the level of the planet you know we act in a level of you know glasses and bottles and C stuff um

and I think that's nice but what I became interested in through processing was that you know we don't act on one level of real of reality or SP temporal scale you know what we're interested in acting on depends on the circumstance and there's lots of nice sort of perceptual psychology experiments but shows the level of detail we perceive also depends on the circumstance um of what our actions are relative to so there are these things called naon figures which I don't I'd be nice to be able to get picture one up actually but basically they're just a shape made up of lots of little smaller shapes yeah yeah yeah you can have like a h made up of little hes right um and there's a general Global precedence effect so people tend to see the bigger letter rather than the letters that make it up unless you present that in the context of asking someone to read a paper right if they're asked to read a paper then they'll they're in reading mode and they're perceiving at the level of the individual letters if you're asking them to identify a figure they're perceiving at the level of the hole um and you get nice effects where this doesn't just happen on a sort of immediate situation basis but based on the kind of actions that you habitually engage in so one of my favorite results and I cannot remember the citation right now it's I think it's Lee the main author her name is Lee um but it's a

paper showing that people who play a lot of Call of Duty of Duty one I like that one dramatically changes I really love that result yeah so people's yes we're sort of perceiving at the rough medium level of the hierarchy but that's going to vary a lot with the kind of things that you do and you know like you can imagine a visual artist's perception is going to be very different for that reason because they are operating perhaps at a finer grain or even perhaps and more cor scin with a good degree of regularity um yep so yeah it was kind of taking the intermediate level view of Consciousness and saying if we've got more than one middle level like is it you know how does that actually shift with action and it makes nice sense of lots of experiments like some sort of intentional blindness type experiments and that kind of thing really nice yeah so this is I can just tell people this is Lee and colleagues 2009 enhancing the contrast sensitivity function through action video game training Nature Neuroscience yeah really cool I mean I I have to take this opportunity to plug myself um funly enough because before we even started speaking it's it's weird I went on this I just wrote this paper called um it's titled currently how preferences enslave attention uh calling into doubt the exogenous endogenous distinction from the perspective of active inference and it's really funny because I try and give a really formal account of exactly this

which is I think that um attention is ultimately governed by the prior that we possess and there's a really but I think what's interesting about this is um the interplay of the preferences mhm so for example I I talk about this in the paper I say um why do I uh you know why am I paying attention to speaking to you right now but then a fluffy dog might walk into the room and suddenly my attention is all geared towards that but then I have this sort of counterveiling preference to go back to speaking to you and then suddenly if the building starts burning I'm going to have absolutely no uh choice but to pay attention to the fire and and to jump out accordingly so it seems like in addition to the high IAL generative model that Maps quite nicely onto a hierarchy of attention and I'm trying I've been trying to uh answer or explain what governs that hierarchy and I'm just curious about I mean I can send you the paper um because it really is very very very I mean amazingly convergent with your paper and I should have read it before and it's my fault but you are now cited so don't worry um because I've been reading obviously your work I've been thinking about so for example why is it that when I'm completely ravenous all I can pay attention to is the fridge and you know the the food around me like the sources of food the potential sources of food and I argue that I think it might be twofold one is that uh those preferences are encoded deeper in the generative

hierarchy than other preferences and I think that the reason why that's important because if you're a predictive processing person that means that the predictions issuing from that deeper level are modulating all of the more superficial levels so I think that I think that stands right because it's like what if there was something above it that it the thing that we're talking about would itself be modulated but my Hunger is never really modulated right like if I'm starving there's not much more that can top down change that so that's one thing the depth the other thing was Precision waiting I think that these preferences right so why when I'm ravenous uh do I go out and get food um and pay attention to food more than anything is because that preference gets extra precision waiting extrinsic gain um I'm curious about just generally what you think about that proposal and whether there's a like there are added Dimensions to this kind of um attentional hierarchy that need to be countenanced I think what I like about it what you how you've just described it in particular is when you think of something like hunger and the kind of significant role it plays it's just one of those nice cases where you have this if you know if you have if you approach this view of like a hierarchical you of cognition from a very maybe classical Cai perspective your view of what's going to be at the top what's going to be the really fundamental thing that's

modulating everything else it's so different from when you approach it through actually you know being a human being in the world who gets hungry yeah realizes that you know the things that modulate everything else you do aren't things like the truths of logic right right

um so yeah I think I like that idea that I think hung is a really good example of where you see that you know your hyper priors or your fundamental priors aren't is purely ration pure like sort of rational detached truths about how the world Works they things about their needs essentially you know that's what's at the root of it all is needs

um and how that relates to all the like attentional switches we actually make in everyday life you know when we're distracted by like a cute animal or like how that emerges out of these really fundamental needs of obviously super hard to draw draw out because it's so so complicated but yeah when you have something like hungry you do see Hunger makes it apparent well because there seem to me to be a uh there's a dynamic element to it so I think uh what I say also very explicit in the paper is that we're not always paying attention to being hungry because if I did I would be obese right like I it would just be really really bad for me um however that so that that's one dimension the other dimension is the that however when I am hungry it takes precedence yeah there's a kind of it's not a paradox but there's

a kind of tension there which is that it doesn't always seem to be occupying super high Precision weight yeah but when it when I have prediction error or free energy with respect to that preference there seems to be a prioritization of the fulfilling of that preference so I I I invoke something like a higher order AC mental action policy which is something like when there is free energy at this level put extra Precision synaptic gain on that preference another interesting idea I could run through you know could run through with you is the idea that I don't think that I think these priors these hyper priors or phenotypic prior whatever you want to call them they might have inherently quite High Precision weight and I'm curious about what you think of this just phenomenologically is it the case I've been trying to think whether it's the case that um in some ways a very phenomenal experience is shaped by these priors in very subtle ways and I think that comes out in the fact that um I may not be aroused or hungry or scared but if there's a loud noise I'm immediately paying attention to it or I might notice uh you know like there's a reason why advertising works right I might find myself getting drawn to a poster of a burger or whatever which implies to me that either this this higher order policy is very sensitive very contextually sensitive and at the slightest contextual cue it will just

deploy High Precision
or there's a kind of standard Baseline
level of precision weight that these
preferences have such that when they
reach over it's very easy for them to
reach over a threshold such that they
grab attention I'm not quite sure how to
unpickle yeah like sort of a low
threshold for Activation rather than
yeah so I'm not I'm just curious about
what you think about I mean I'm curious
really fundamentally about what you
think about the phenomenological idea
that our
experience is permeated by those
preferences right our attention in many
ways is permeated by those preferences
is that the case or can you imagine sort
of is it the case that if I am well
sated and so
on Hunger as a or or satiation just
basically doesn't affect affect my
cognition whatsoever what is there a
kind of what's the kind of blurry level
of influence do you think that those
preferences
have yeah I that's a really interesting
point like to what extent is your
phenomenology shaped by your needs when
your needs are satisfied right right
um I don't know I'm not sure how to
think I think that's I think it's an
interesting question I haven't thought
about the V visual phenomenology stuff
in so long whenever anyone asks me a
question about it there's that tendency
to sort of go and be like what is my
visual phenomenology um I mean I mean if

anyone's listening if anyone knows
whether Merlo Ponte or whoever I'm sure
someone spoken about this um so I think
maybe one aspect is I
suppose maybe what I would say is I
think that we those are like going to be
you know your basic fic needs but I
think as you become a human being as you
become a person this is my M Pony is not
great but I know he talks about some
stuff as you become a person you develop
things that take on the same kind of
status as these more basic fpic needs
yeah I suppose what I would say is I
don't think there's ever a situation
where all of your needs are satiated
yeah because you all you develop other
needs that you know they don't initially
take on the level of a need so like you
know a desire
to have your parent maybe desire to have
your parents approval is probably right
probably want
deep but you can you can pick up you
know new things that initially aren't
need but they become so integral to your
identity that they become need and then
those needs are shaping your perception
in a way that when hunger isn't shaping
your perception that's not to say that
no needs are it's just to say that a
different need takes
salience um and I think this is a really
nice this idea that's what beautiful to
me about this idea of need as being
fundamental need and ident like
constitutive needs it's that it's very
amable to being found both at the

biological level and at the social level
so there's a philosophical Christine
cosgard who has a little of work on this
idea of
self-constitution that as humans we
constitute our identities and they that
gives us needs that we have to act upon
to constitute ourselves and it's very
Canan but it's very sort of rationalist
I mean she's interested in animals but
is generally described in quite
rationalistic human conscious terms but
it's the same kind of thing going on
there as when you're hungry and you need
to eat when
I'm not
um acting in a way that's consistent
with with
my my own view of my
identity there's like a need there that
I'm trying to fulfill in failing right
the same way as when I'm hungry and I'm
not able to get food there's a need
there and I'm trying to fill in failing
yeah I mean I guess that too far off Bas
but no no no I think that's absolutely
right I think just thinking about it you
know even
now I think there's what I I wonder
whether there's always a sort of trickle
back to some overarching hyper prior
right so uh I want to write a great
paper because I want to impress my
parents well it's kind of Aristotelian
in it way well why do I want to please
my parents well because I want I don't
know I mean why would one want to e so
they don't I don't know evolutionarily

so they don't throw me in the bin or
whatever um and so on and so forth and I
was just wonder whether that gets where
that gets bottomed out but I think
that's also what I mean by this idea
that these Hyo priors permeate our lived
experience because even if this is
actually just an instrumental
description I wouldn't say all of all
Behavior or all perception can be traced
back to a a limited set of hyper prior
but I think some uh some can right so
some very superficial goals or seemingly
superficial goals like finishing a paper
yeah have the capacity to be really
embedded within a deeper preference of
being happy right ultimately I guess the
only question there would
be h
hunger something like hunger or more
phenotypic needs they almost seem less
like that because they don't seem to
engender this kind of deep hierarchy
right there don't seem to be plenty of
things I do because they indirectly
contribute to my so
Society um they almost seem like a
different type of Prior such that
they need to be fulfilled but they're
almost singular in the fact that
fulfillment can only come through like
particular action policies anyway this
is super off I'm just sort of but I
guess they're interesting because
they're both very basic and very
specific yeah like hunger as a need is
very fundamental but also there are very
specific things that can fill it it's

not very general whereas if you take a more basic need like I need to have a coherent identity I need to have a coherent personal that's everywhere right that's just everywhere yeah and there's so many different ways I could fulfill that so you might be able to trace back my having this conversation with you about my philosophical work work to a need to have a coherent practical identity but you can't get from the need to have a coherent practical identity to this particular manifestation of it right it's very underspecified

um there's a really nice I think it's it's idea of like the the deductive importance of this kind of idea of an identity so it constrains what you can do but in such a limited way that there's so many different possibilities that they're compatible with and what's interesting about hunger and these more phenotypic needs like I think is that they are really fundamental but also a lot more specific you know you can derive what I must do to fulfill it um so I think that would be a difference yeah no that's interesting ah this going to be fun editing um that's all right that's what academic Academia for cool well there's there's I think just on that point another thing I think is interesting as well is when you think of I think you get this view where some of our needs are really manifestation of social leads so for a society to work it needs to be

constru made you could argue it needs to be made out of individuals that have coherent practical identities like if we don't have coherent identities if what I'm doing now is completely and utterly unrelated to what I'll do tomorrow then you have no ability to interact with me meaningfully right right because I can't make any commitments to you that's really yeah that's really interesting actually there's a sense there in which you can see why the need for my need for coherent practical identity so by which I mean just that you know being a sort of predictable person over time who's committed to certain projects and follows certain narratives if I don't have that then that's not just necessary for me but that's necessary for us as a whole and it becomes a very primitive need of a group of people which I think is interesting do you think that says anything interesting about sort of um it comes back to constraints in many ways yeah that um do you feel like there's one or or you know we think perhaps of the kind of more like the the hyper you know Aristotle would thought of a hyperv if he had that language as um udonia right and in the case of humans it would be it would be reason exercising our reason and and the way that you say well having a fixed Identity or having a set Identity or stable identity is Downstream on me being a functioning member of society and then if you ask the Marxist it would be because that

affords you the opportunity to you know
uh be a be a a capitalist slave or
something I don't know
um but you know they might say Well in
that way you're just a kind of your
preferences are just a vehicle for a
broader political sociopolitical
structures preferences and that might be
ground and then you know what's the
authority over that well is there is
there is there a point I mean I guess my
question is there a point where those
preferences ground out or is it
philosophically vague as to how
recursively you know recursive that can
go yeah and I think so for me I think
where those preferences ground out
they in terms of scale you can have them
being instantiated at lots of different
scales that can be the preference of a
society and the preference of
individuals within that Society but what
they ground out in is this idea of the
ability to
coherently constitute yourself over time
because in as far as your identity isn't
something that you can take as a given
it's not just that you know like a rock
you'll just continue to exist whether or
not you do anything you have to
continuously do something to you have to
create yourself into the future in that
way
whether I exist in the future depends on
Me My Success
now um the con the ultimate constraints
are going to be there are things that I
can't do you know there are things that

if I do them now I won't continue to
exist in the future and there are things
that I have to do to continue to exist
in the future but in the space of things
that I have to do to continue existing
the possibilities are pretty vast um
there's tons and tons of different ways
I could continue my existence so from
the fact that I have to
continually construct this continuous
self process that tells you very little
about how I have to do it it tells you
things that I can't do but
it gives you very little Traction in
pinning
down what I will do at each moment in
time so in as far as there are these
kind of ultimate constraints they don't
get you super far in pinning
constraining any particular individual
yeah yeah yeah yeah well there's always
that F you know there is always that
individual versus Collective or that
that that that dialectic right like and
I guess that is that's imminent in an
active inference more fundamentally
right like unless yeah I don't think any
of us have this kind of notion of a
process ontology where it's totally
loose that you could be anything um yeah
I don't want to go back to that because
we've I don't think there's an answer um
talking about grounding out and
bottoming out I want to go back if I may
to this notion of intermediate level
processing so so perception if I've you
know read these papers correctly
perception in many ways is is sort of

the guide of action because it allows
you in phenomenal
space that which is emerges in
phenomenal space is that which you can
act on let's say right so it kind of
very JJ Gibson idea um but the reason
why you know I don't perceive the uh the
atoms of my phone is because I can't
grasp ATS right so there's this nice
Mutual coupling and that runs down all
the way down to the phenomenology but as
you say very nicely in your paper if I
take a brick wall I think this is a
celian idea if I have a sort of brick
wall there isn't one level of which I
can perceive that brick wall uh I can
stand away from it such that it becomes
a kind of uh ambiguous mass of brick or
something but I can't see the individual
bricks uh or I can go even closer and
you know this is a illusion of the cigar
or whatever and I can I can Noti I can
then lookig and there's kind of this
fractal nature of perception so we're
talking about grounding out and boming
out is there you you speak about this a
little bit in your paper but I'm curious
about what you think now is there a
point at which perception bottoms out um
and can is that again just mappable to
to action such that it grounds out at
the point where I can't act right I can
always grab the cigar I can always I
don't know what I could do about with
the sort of brick wall but in principle
I could do something with it is there a
yeah where does that where does that
chain

stop yeah I will I just need to get some
water sorry yeah yeah yeah no worri
sorry about that no wor so yeah what is
perception ground at what level
um I mean I know there's experiments you
know you can look at all the sort of
stuff on what is the actual capability
of the sensors that we have um in terms
of people's ability to kind of
discriminate the presence or absence of
a single Photon and that kind of thing
um but in terms of
phenomenology I don't know I don't know
whether I'm trying to remember what I
said in the paper because it's
been would you like to know I would like
to know was like six
years um while I'm looking you can you
can try you can try and remember
um but no you do you do mention it that
I mean it it's I think I think it's
totally reasonable to say I I don't
think there's anything intuitively wrong
with a notion that it will bottom out at
the point of uh attent you know
actionable inaccessibility if that makes
sense right like it just doesn't make
sense given the body that I have and
again it's very sort of Merlo ponan idea
that it just doesn't make sense to
perceive atoms with a hand that is uh
manipulable only kind of quite course
levels right like I can't grip just
because of the nature of the hand that I
have I can't grip um atoms which makes
me think I it's a different
question there's a uh I think again
intuitively we think there's a kind of

universality of perception because we're
our sort of folk science or folk
philosophy is telling us that we just
receive the world and we recapitulate it
perfectly a kind of direct realist um
way of viewing things um to what degree
does your argument that perception is
kind of affording action in the sense
that it gives us this intermediate level
perception that's grippable and whatever
to what degree is that kind of a sa WF
uh account of vision I mean by that I
mean to what degree I mean how much does
affordance and action relativize and
permeate perception is there anything
beyond it um that that's kind of what I
I mean I don't think any you know I
really J I don't know I might get shout
that I'm not sure Gibson really tells us
yeah so I I think for
me I would say no in the sense in or in
the perception of
objects the perception of an object I
think is always cashed out in terms of
action possibilities any time you
perceive anything as an object you
perceive something that you can pick up
that you can rotate that has other sides
to
it what I'm I think the reason I'm not
entirely sure whether that's exhaustive
of perception as a whole is in terms of
kind of sort of Altered States Of
Consciousness type experience where you
have perceptions that aren't really
perceptions of objects anymore um you
know so when you
have I don't know whether it's some of

the extreme states that people apparently get from meditation or from LSD and you have that kind of very non-ordinary visual phenomenology I think there it's very hard to see what the action is um behind those kind of experiences I mean the closest I've had to one um is playing around with you that Lucille like thing the dream machine thing and you get those kind of you know swirling mandalas and all that kind of thing yeah yeah yeah um and it's not really clear what the ACT it's not clear what the action is there that's very nice that's a good idea there is also the sense at least though with those experiences right like you know how they'll change if you move your your I think you do in my experience you have an awareness of how you know if I move my eyes this way the I'm going to get that sense the color shifting across my visual field this way there's still some kind of contingencies to them what there isn't obviously is any sort of sense of any kind of needs of like why should I act this way or that way you know I can actually change it but there's no relevance of that kind of visual phenomenology to anything I might need or want in the same way that there is with like tables chairs lamps and water bottles yeah yeah yeah yeah that's that's a really that's a great paper idea by the way for anyone listening You' just got given a great paper idea to what degree what are the affordances in psychedelics I mean again I haven't

thought about visual phenomenology in
ages and it's it's fun to think about
yeah no it's cool um I mean it makes me
think you know we always think of
affordances in terms of physical actions
but the nice bit of active inference is
people like Ian SED Smith and others is
uh the alignment let's say of mental
actions and physical actions such that
they can be described via the same
formalisms and I wonder whether just
really FR this you know it's in the dark
I haven't thought about it whether the
affordances there might be mental
affordances in psychedelic States
especially at this very deep level of
let's say like a DMT trip or non-dual
awareness um but again that said it's
still that still foregrounds the notion
of needs and uh a kind of autonomous
being who has needs onto themselves
because that is only the that's the kind
of the only way that affordances make
sense

so you know I was going to ask you there
you you say in one of your this I think
this will be my final question because
it's a big one well it's not actually a
big one but it's um you know about it's
comes back to the self I'm always going
to come back to the self in many ways
because again that's what I've been
writing about and that's what I'm most
interested in um you say you don't
really go into this idea but I think
it's really important that you say and I
wonder what you talk about in the book
as well you say in um your 2022 paper

expecting some action having a temporarily fit counterfactually deep self model seems an obvious prerequisite for the presence of high all to self-consciousness but it is not immediately clear why it would be necessary for the emergence of more basic phenomenal properties such as the immediate visual experience of an apple on the table and then I think you go on to talk about um the this kind of action based account of visual perception that does give you the visual experience of an apple on table but I wanted to sort of when I first read that I was thinking yeah it doesn't give you it certainly an apple on a table but it certainly doesn't also give you everything that we can talk about in terms of selfhood so I've been very interested in uh what the phenomenologist called pre-reflective self-awareness um or what people also refer to as the sense of self as subject so this is the kind of you can you can be very deflation about it and just say um the subjectivity that's imminent in all conscious experience or the perspectival or the very sort of non-conceptual possessiveness that seems to accompany all or not even accompany it embed in all conscious experience and here I'm thinking about people like Sean Gallagher and Dan Sarvey and all about Hle and M. Ponte and I'm really really fascinated than them but I'm wondering whether you thought about that as well in terms of what act of inference can say about the self um there's a nice

paper by Matas wasak who talks about the
I versus me and the fact that act of
inference in principle can give you very
much the uh the me the self as OB what I
take is the self the sense of the self
as object but it can't really give you
the why which is the sense of self as
subject um I'm wondering whether you
thought about that at all in terms of uh
Act of inference and the phenomenology I
know you focus more on visual
phenomenology but underlying all visual
phenomenology is this kind of
subjectivity this fact that there is a
for meanus of visual perception so I'm
just wondering whether you'd given that
any any consideration because you did
make that comment in in that paper I I
don't know the paper that you just
mentioned I'd be interested to so I
don't I'm talking about so you made the
comment about the fact that WN so yes so
wnac is I think it was in Frontiers
um I can just for the AUD I and me the
oh it's not is it I and me the self in
the context of Consciousness Frontiers
in Psychology 2018 I can send that to
you yeah it's interesting
um so sorry the question was more was on
well the question really feeds off this
idea that you said that the this notion
of deep uh temporal counterfactual uh
planning in many ways affords a kind of
uh narrative self a kind of Rich self
model that's capable of being reflected
on right like I am a thing that causes
things in the world this idea that we
are a hidden cause we are the best

guests of uh multimodal sensory
information and so on and all these
things that George is writing about and
my response to George has always been
well yeah that's very good for the
self's object and by that I mean the
self that can be reflected on so I am
Darius I'm 25 I'm from London and so on
yeah like that implies I'm a rified
thing that uh it can be cast as a latent
cause and I think active inference gives
you that what I'm asking is you said
well it's not enough to give you this
kind of very basil visual perception of
course I mean I think that's kind of
clear what I'm also saying is it it
doesn't just give not give you uh visual
perception per se it also doesn't give
you the subjectivity or the very
non-conceptual sense of self imminent in
visual perception okay yeah so I'm just
curious about what you think about that
yeah and I think that I think that's
maybe tied to this way in which I think
the sense of
self the the way I think of a self being
very different from the way active
inference necessarily does which is a
sense of sort of a kind of intrinsic
striving I suppose in the sense that
it's not just I have this model and if
something happens I need to respond to
get rid of disruption it's that I am and
also maybe to what we said earlier about
this idea I think that there's never a
position when all of your needs are
satiated is always some sort of need
it's just which one is being prioritized

at which point in time so the basic
sense of self I think that is
underpinning visual perception is going
to be that sense
of imbalance
or misachi super nice I about that yeah
I mean that that again is kind of quite
punching I guess but like um the B the
basic self is like misalignment Mis
achievement that we're trying to align
and that's always a futile process
because you can never get rid of all of
this mis achievement you can just get
rid of one element of it and something
else becomes more Salient and then the
kind of reflective self is what allows
you perhaps to you know that's when you
get into planning about it and all of
that sort of stuff and perhaps in order
to plan you need to act as though your
needs are going to be the same so even
though time in ways that you know you
might there's no necessity behind your
needs staying the same but if you want
to make plans in the world you're going
to have to act as if they will in order
to project yourself into the future um
oh that's a good question so a good
question as to whether you can
fake alternate
desires that's a different question yeah
that is really nice I mean I'm curious
whether you think there's kind of um
this
nonreflective
uh precarious
self-awareness is do you think that man
what does that manifest as in your IDE

in this idea is it conscious is sorry is
it um affect or is it even deeper than
affect yeah so for me it's what it is it
effect it's it's probably effect whether
whether it's Consciousness I think I'm
undecided on whether you need some sort
of reflective capacity for it to be
conscious so I would say you know every
living system has this like basic
striving and need a bacterium has it
anything that is a metabolically a
metabolic cycle has that um which if you
say that that as soon as you have that
you have you know something like
Consciousness you end up with a very
general notion of it um and I haven't
really got to what I
think I would be willing
to I think it's less crazy to me um if
you think that that's what it is is to
be to have purposiveness and
intentionality then you might be
ascribed and you think that that's what
is to have Consciousness you might be
ascribing it to very very simple life
forms the main thing for me is that
you're not ascribing it to
everything that that very minimal notion
of striving is very general and weak
that would be okay for me for that to be
something like some sort of minimal
phenomenality yeah yeah yeah yeah just
the main thing I think is the most
important thing to avoid if we want
concept if they we want there to be to
be realists about the mind about
phenomenality is I feel that once that
becomes a property of everything it

becomes a less important or real property so even if it's watered down to the extent that we can say that there's something that bacteria have effective States that's still okay as long as um and tables don't Right From perspective yes well rather you than me to draw that dividing line um but I'm sure you I'm sure one can I mean I think just to sort of say what I'm thinking currently I don't think that this form of Consciousness needs to be the product of any meta reflection there are people that think that and it's a very sort of these are not phenomenologists per se these are something more like um analytical philosophy's mind so people like Armstrong like and Rosen far I've been thinking about this because they argue that sort of for this this intransitive Consciousness can only come about because it's the object of a higher order mental state yeah but then I mean this is such a because it's obviously leads to an infinite regress which is like is that mental state conscious and if so how is it conscious without itself being the object and so on so I I don't know I'm I'm sympathetic again I'm not going to stake any solid claims I'm sympathetic to the idea that there is just minimal phenomenality that doesn't need uh that doesn't rest on any higher order cognitive function MH and you know is the fact that there is this uh at least plausible interiority about

living things that autonomous or
adaptive
things does that feed into the minimal
phenomenality is that the ground of
minimal phenomenality such that it's
kind of darine and it's this kind of
imminent anxiety haiger might have even
said you know that this future striving
that is consciousness I don't know but I
don't hate it actually um there are so
many people who are trying to get to the
BM of this what is this minimal
phenomenality whether it's affect or uh
whether it's salience or whether it's
time or whatever that um it's hard to
say but that that is a lot to there a
lot to chew on I'm also aware of the
time yeah I want to say just in terms of
the reflection thing if that's what
you're thinking about and you're not
convinced by There's A really lovely
book that I just came across recently
called on reflection um I again first
named um but it's corn BL I want to say
Hillary corn BL and it's literally going
through how philosophers have relied
upon reflection the idea of reflection
to get themselves out all of these
problems with their first order theories
we can't explain it in terms of this but
if you reflect on it yeah yeah yeah then
you have a self or then you have this
and he's like you know if you haven't
got the first sort of thing that you're
reflecting on sorted then just moving to
reflection is just passing the buck yeah
of course it's a nice short book but he
I think it is a really common move to

sort of rely upon some sort of higher
order reflection to explain how
something you can't explain happens
right um so yeah I I I think that's
maybe why I'm more interested in minimal
phenomenality because I think there's no
point talking about everything else
until you figured out what that thing is
and then you need can sort of explain
how you can have reflection on it yeah
I'd really recommend it yeah that's
that's cool I think yeah that is cool I
think the idea they would have is that I
mean just in terms of this very sort of
fact this idea that minimal
phenomenality is itself Downstream on
reflection on minimal phenomenality from
some other mental state is that that
other mental state that higher older
mental state is not conscious MH but
then you just have this issue of how is
a non-conscious mental state intending a
conscious mental state anyway this for
another day Kate this was really fun I
mean a complete uh mind frazz and brain
roller coaster but really amazing and
insightful so I'm extremely grateful
genuinely um I say it to everyone I I
hope it doesn't end up seeming
deflationary or or boring or as long as
you get their names right when you say
it that's the main thing I do try and
get people's names right I do try and
get people's names right I'm always like
scrolling away to make sure I get the
paper names right and whatnot but um
genuinely so much to think about not
only I'm sure in terms of my own uh case

because I have now a lot to think about
but hopefully in terms of the audience
as well um I always like to give people
just a little bit at the end just talk
about where they can be found on the
internet
presumably rather than physical location
because that would be weird um but yeah
in terms of Twitter or email or whatever
uh and also what kind of work they have
upcoming that people can look forward to
cool yeah and thanks so much for having
me this is the first time I've done any
sort of extended I used to be the I used
to be a journalist so I used to be the
person interviewing people so it's been
really interesting being on the other
end of the interview oh well you you
smashed it honestly that it was nice it
was a nice like chill first interview um
yeah so I'm on Twitter I'm trying not to
be as much but I am on there just as
Katherine Nave um just all one
word in terms of upcoming work there's
the book coming out hopefully at the end
of this year I'm thinking it's still it
should be maybe by October but it could
be February it's been in the works for a
long time how long is the book may I ask
oh I not sure I think I want to say
like I don't know just just to warn
people I guess I think
not yeah maybe like 80,000 words yeah I
think it's about 70 or 80,000 words
sounds like a lot yeah but there's an
appendix yeah yeah yeah and it's funny
it's I've been told it's funny and this
was built off your PhD you said yeah

okay nice yeah um yeah it's supposed to be relatively

accessible I would say like certainly you can you can dive into it and get what you want out of it so like the first half is probably I think is like a you know you think is a relatively decent introduction to the kind of basic philosophy of the fry principle and then the second half is more if you're interested in stuff like purposiveness and intentionality and an active concept you know more oriented towards that um there's also some blog posts I've got a post on dialectical systems which kind of summarizes some of the arguments of the book um and I've got a little substack post I don't have a substack I just have a single post that kind of tries to give an overview of the book's main arguments so those might be worth checking out

um and the next thing I'm working on is a big paper about Hegel so that'll be a while but yeah well thank you again so much from me um and yeah and and from The Institute as well I'm sure everyone's going to love it and and everyone should go get the book when it's out of course yeah thanks for inviting me I really enjoyed talking to you about it all right no absolute pleasure all right thank you everyone for watching as well byebye by